

1 **Feasibility of Field-Derived High-Resolution Event Replay for Traffic-**
2 **Signal Controller Testing**

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2 **Objectives.** Manual controller testing cannot reproduce everything a signal controller experiences over a day, including
3 interacting vehicle, pedestrian, coordination, and preemption activity. This study evaluates whether recorded high-
4 resolution field events can be replayed to test controllers and their output sequences automatically aligned and
5 compared to identify operational differences.

6 **Methods.** An open-source Python package converts high-resolution logs to National Transportation Communications
7 for Intelligent Transportation Systems Protocol (NTCIP) calls and replays them to test controllers. Output events are
8 grouped by timestamp and aligned using dynamic time warping with Jaccard distance between event sets. It also
9 checks clearance intervals and incompatible signal phases and overlaps. Approximately 23 hours of events from each
10 of 25 production configurations were replayed to controller emulators. Separate runs tested repeatability, known timing
11 changes, and practical use for software-version acceptance.

12 **Findings.** Controller outputs from separate unchanged runs aligned with 99.6 percent mean sequence match and 97.7
13 percent mean timing match, demonstrating that controller responses to field-derived replay were deterministic enough
14 for automatic comparison. The method flagged operational differences in 11 of 14 configurations with deliberate
15 overlap clearance-setting changes and none of the 11 unchanged configurations. Applied to a software update, it iden-
16 tified differences within similar full-day operation, including correction of a rail-preemption exit bug. No incompatible
17 signal phases or overlaps were detected.

18 **Novelty.** The study combines replay of full-day event sequences actually experienced by live field controllers with
19 automatic alignment and comparison of complete controller-output sequences across numerous production timing
20 configurations.

21 **Practical Applications.** The open-source workflow expands acceptance testing beyond manually toggling individual
22 inputs. Agencies can replay signal activity across many configurations, identify unexpected differences before field
23 deployment, reproduce field failures, and test software or timing corrections against the same inputs. The Oregon
24 Department of Transportation (ODOT) now uses the workflow for software-version acceptance.

1 INTRODUCTION

2 Controller acceptance testing is commonly limited by staff time and by the inputs staff can reasonably operate by
3 hand. Manually toggling detector, pedestrian, and preemption inputs remains useful for checking individual functions,
4 but it cannot reproduce a full day of interacting calls, coordination transitions, and preemption activity across many
5 production timing configurations.

6 This project uses field operation as the test sequence. Approximately 23 hours of high-resolution events actually expe-
7 rienced by each of 25 live field controllers are converted to National Transportation Communications for Intelligent
8 Transportation Systems Protocol (NTCIP) calls and replayed to test controllers. The resulting new output events are
9 compared with the original field-derived baseline. The central question is:

10 Can high-resolution field events be replayed to test controllers, and can the resulting output sequences be aligned
11 and compared automatically to identify operational differences?

12 Feasibility requires both repeatability and sensitivity to change. Replaying the same inputs must produce controller
13 outputs similar enough to align reliably, while actual changes in signal operation must remain visible. The comparison
14 must also direct staff to important differences in signal phases, clearances, pedestrian service, preemption, coordina-
15 tion, and incompatible movements without requiring them to inspect millions of event records.

16 The evaluation covers 25 production configurations and three emulator campaigns using MAXTIME controller
17 software: version 2.15.1, version 2.18.1, and version 2.18.1 after overlap trailing yellow and red clearance settings
18 were changed to prevent occasional truncated yellow and red intervals. The primary contribution is evidence that
19 field-derived controller outputs were repeatable enough for automated comparison and that the method successfully
20 identified real operational differences. The specific software versions provide the case study; the finding is that this
21 form of testing works.

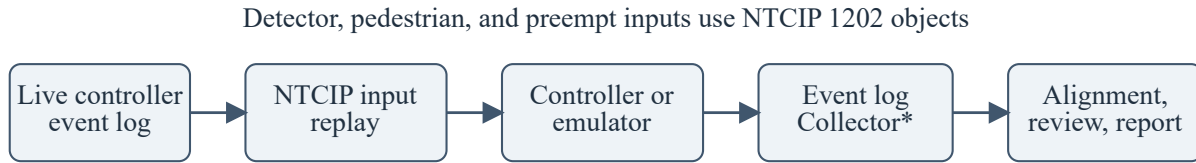
22 RELATED WORK

23 Li et al. developed a controller automated-testing tool using its CIDScript language (Li et al. 2008). A subsequent
24 Idaho Transportation Department system used Extensible Markup Language (XML) scripts to activate inputs and
25 verify predefined responses (Ahmed et al. 2010). Tung first demonstrated automated testing on a National Electrical
26 Manufacturers Association (NEMA) TS2 Type-1 controller (Tung 2012), then reported 20 NTCIP-based test programs
27 evaluated on five compliant controller models (Tung 2015). These systems established the value of repeatable
28 automated tests. The present method differs by replaying real events recorded as they occurred at live field controllers
29 and using a prior complete output trace as the expected response rather than requiring staff to author every input and
30 assertion.

31 Controller-in-the-loop studies have connected controllers and emulators to traffic simulation through physical or
32 virtual interfaces (Wang et al. 2021). Stevanovic, Klanac, and Radivojevic evaluated high-resolution logging across
33 six controller vendors (Stevanovic et al. 2017). Those studies support the use of controlled controller environments
34 and show that output-event semantics require platform-specific validation. They did not evaluate field-log replay for
35 software regression across production configurations.

36 High-resolution event records provide tenth-second controller and detector histories (Sturdevant et al. 2012). Dynamic
37 time warping (DTW) provides a monotone alignment for sequences with local timing variation (Sakoe and Chiba
38 1978). The literature reviewed for this study did not identify a prior evaluation combining full-day replay of events
39 recorded from live field controllers, automatic complete-output alignment, and repeated software and timing tests
40 across numerous production configurations.

1 REPLAY AND TEST ARCHITECTURE



*The current collector reads MAXTIME event logs.
It can be extended to other controller types.

Figure 1: Field-derived testing workflow. Replay inputs use NTCIP 1202 objects. The current event-log collector reads MAXTIME logs and can be extended to other controller types.

2 Field event conversion

Each of the 25 source logs spans approximately 9:00 a.m. to 8:00 a.m. the next day. Together they contain 7,296,218 records: approximately 23 hours per controller, or 575 hours when summed across all controllers. Replay selects vehicle detector, pedestrian detector, and preemption changes. Table 1 shows the conversion implemented in the package.

Input	Event codes	Parameter	NTCIP replay state
Vehicle detector	81 off; 82 on	Detector 1–64	Update one bit in an eight-detector group and write the cumulative integer.
Pedestrian detector	89 off; 90 on	Detector 1–64	Update one bit in an eight-detector group and write the cumulative integer.
Preemption	104 off; 102 on	Preempt number	Write the corresponding Boolean preempt state.

Table 1. Conversion from field events to replay inputs.

Repeated on/off records are repaired by imputing the missing opposite transition. The Oregon Department of Transportation (ODOT) reserves detector numbers 65 and above for nonstandard uses, so those detector numbers are excluded from replay. Before replay, all calls are reset to zero. The package then sends Simple Network Management Protocol (SNMP) SET operations to NTCIP 1202 vehicle, pedestrian, and preempt objects at the recorded time of day (American Association of State Highway and Transportation Officials and Institute of Transportation Engineers and National Electrical Manufacturers Association 2023). Preserving time of day also exercises coordination and time-of-day plan changes.

The evaluated implementation used MAXTIME emulators loaded with production databases. Emulator accommodations removed detector delay/extension, disabled unused input/output modules, removed inverted preempt logic, used localhost peer addresses, and mapped overlap pedestrian calls to reserved detector inputs because MAXTIME was not accepting overlap pedestrian calls through NTCIP. Output events were collected through a MAXTIME Hypertext Transfer Protocol (HTTP) interface. Because replay inputs are sent through NTCIP 1202 objects, the input side is not tied to MAXTIME or another controller brand. The event-log collector currently reads MAXTIME logs; it can be extended to another controller family by mapping that controller’s events to the common timestamp/event/parameter format.

Parallel replay and latency control

Signals assigned to available emulator targets run concurrently. The 25 configurations were processed in batches over approximately one week without requiring staff to operate each input. After the initial setup, future releases are expected to require only a few hours of staff preparation and review; this is operational experience, not a formal labor study.

The workstation clock was found to drift relative to the test-controller clocks during the long replays. Without correction, the same input could gradually be recorded at a different controller time. The replay therefore schedules each command early by a per-controller latency offset. At periodic intervals, the package matches isolated detector-on inputs with the corresponding detector-on events recorded by the controller, compares the scheduled send time with the controller timestamp, and updates the offset to the measured median delay. The offset changes gradually during playback to avoid an abrupt timing jump. The MAXTIME version 2.18.1 release database contains 707,458 latency

1 samples and 6,577 applied updates. Each campaign contains 627,240 replay commands and between 7.57 and 8.02
2 million controller-output events.

3 OUTPUT ALIGNMENT AND AUTOMATED CHECKS

4 Discrete-event dynamic time warping

5 Events recorded at the same time, or within a 0.25-second grouping window, are represented as one set of (event
6 identifier, parameter) pairs. This prevents the arbitrary database order of simultaneous events from affecting the result.
7 The timestamps define the groups and support the initial time alignment, but they are not part of the DTW sequence
8 cost. Let A_i and B_j be the i th and j th event groups in the reference and candidate sequences. Their local sequence cost
9 is Jaccard distance:

$$d(A_i, B_j) = 1 - \frac{|A_i \cap B_j|}{|A_i \cup B_j|}.$$

10 Identical event groups cost zero. Missing or additional events increase the cost. The cumulative DTW cost is

$$D(i, j) = d(A_i, B_j) + \min(D(i-1, j), D(i, j-1), D(i-1, j-1)),$$

11 with $D(0, 0) = 0$ and inaccessible borders set to infinity. Backtracking produces an order-preserving path based on
12 event content. Horizontal and vertical path segments identify additional or missing event groups. After this sequence
13 alignment is established, the timing comparison separately evaluates the controller timestamps of event groups that
14 match exactly.

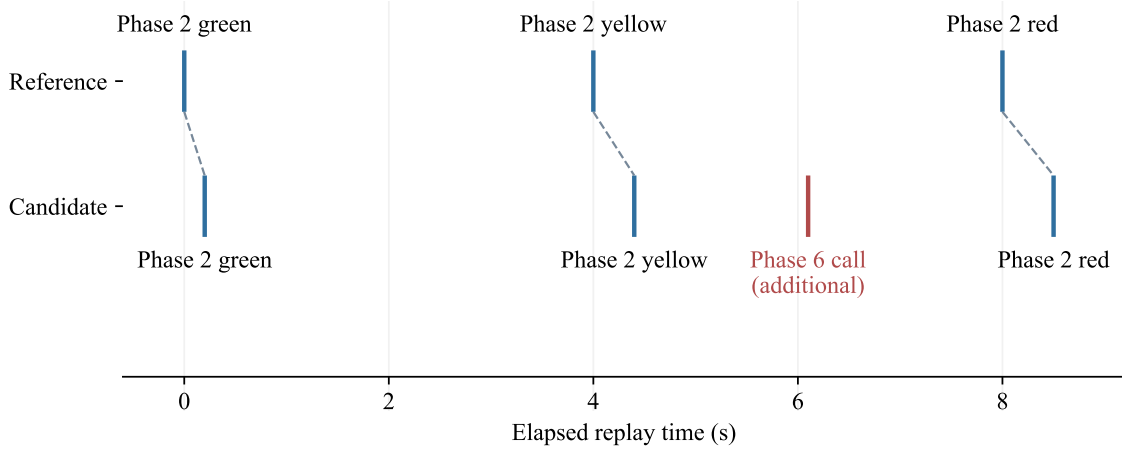


Figure 2: Simplified event-sequence alignment. DTW aligns matching event groups by content and flags the additional Phase 6 call. After the sequence alignment is established, timestamp differences are evaluated separately.

15 Sequence scores are calculated in 45-minute windows advanced every 40 minutes. Sixty seconds are clipped from each
16 window edge to avoid penalizing a cycle split at a boundary. Sequence match is the percentage of aligned event-group
17 pairs with zero Jaccard distance. Timing match is the percentage of exactly matched groups within 0.50 seconds after
18 removing the remaining median clock offset from their timestamp differences. A configuration passes at 95 percent
19 sequence match and 90 percent timing match.

20 Vehicle phase-call similarity provides a separate replay-reliability check. Windows below 85 percent are displayed but
21 excluded from the configuration average. This helps distinguish controller differences from a test environment that
22 did not reproduce the intended inputs.

23 Operational checks and virtual conflict monitor

24 The automated review is broader than sequence scoring. It compares phase and overlap intervals, clearance durations,
25 pedestrian service, preemption, and coordination-transition states. It highlights short or irregular yellow/red clearances
26 and material changes in occurrence or duration.

27 A software-based virtual conflict monitor reconstructs active phase, overlap, pedestrian, and overlap-pedestrian states
28 from every collected output event. Each state is checked against configuration-specific incompatible pairs. The study

1 definitions contain 863 incompatible pairs across 21 configurations. No conflicts were stored in the MAXTIME 2.15.1,
2 MAXTIME 2.18.1, or modified-overlap-setting campaigns. This automated check is a major increase in test coverage,
3 but it complements rather than replaces a cabinet conflict monitor or formal safety certification.

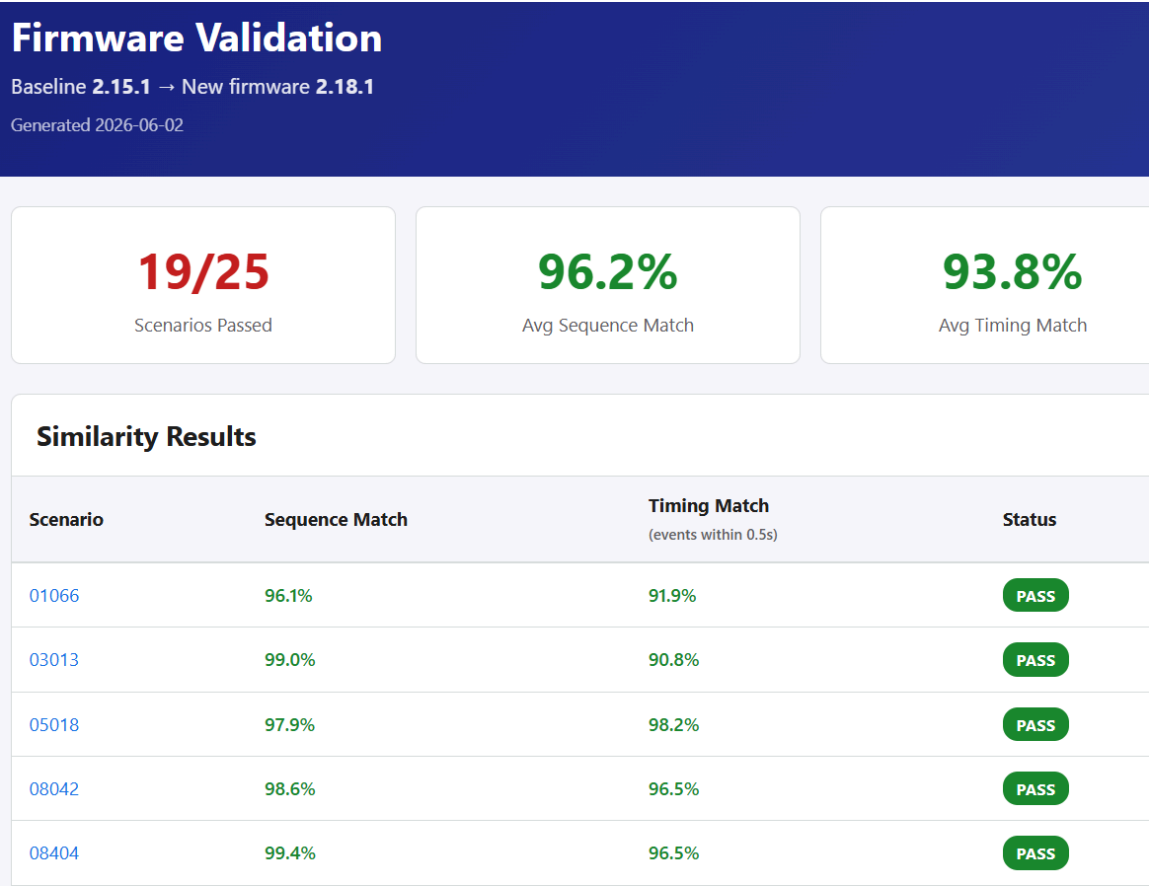


Figure 3: Example automated report. Aggregate sequence and timing results lead to focused views of clearance, pedestrian, preemption, transition, and conflict findings, allowing staff to review exceptions instead of the full event stream.

1 EXPERIMENTAL DESIGN

2 Table 2 separates the two tests. The release comparison represents normal acceptance testing. The same-software
3 comparison provides a repeatability group and a known operational intervention.

Test	Reference	Candidate	Configurations	Question
Software release	MAXTIME 2.15.1	MAXTIME 2.18.1	25	Can full-day outputs be aligned and meaningful release differences surfaced?
Overlap clearance settings	MAXTIME 2.18.1	Same software with applicable overlap trail settings	14 changed; 11 unchanged	Are unchanged replays repeatable, and are deliberate changes detected?

4 Table 2. Evaluation design.

5 Occasional truncated overlap yellow and red clearance intervals had been observed. To prevent them, trail-yellow
6 settings were added to applicable overlaps, with trail-red and red-revert settings changed as needed. Transaction
7 histories identified these edits in 14 configurations; 11 had no relevant edit. The test provides a known operational
8 difference while holding the software version constant. A changed setting is detectable only if the recorded day
9 exercises the affected overlap operation.

10 RESULTS

11 Field replay produced repeatable controller outputs

12 All 11 unchanged configurations passed across separate MAXTIME 2.18.1 replay campaigns. Their mean sequence
13 match was 99.6 percent (range 99.1–100.0 percent), and their mean timing match was 97.7 percent (range 90.6–99.7
14 percent). These were independent full-day controller executions, not a recorded output compared with itself. This is
15 the principal feasibility result: the controller responses were deterministic enough for DTW to align and compare them
16 automatically.

17 Eleven of 14 configurations with applicable overlap clearance-setting changes were flagged, while none of the 11
18 unchanged configurations was flagged. The three changed configurations that passed had sequence matches of 99.3,
19 99.9, and 100.0 percent; the recorded input apparently did not materially exercise the affected behavior. Table 3
20 summarizes the result without implying exhaustive coverage of every possible timing change.

Group	Flagged	Passed	Total
Applicable overlap clearance edit	11	3	14
No relevant edit	0	11	11

21 Table 3. Same-software repeatability and parameter-intervention results.

22 The overlap-setting test was motivated by the type of event shown in Figure 4. The automated check detected a 3.0-
23 second overlap yellow interval, below ODOT’s 3.5-second minimum review threshold and the 4.0-second median
24 yellow interval for that signal phase. It occurred with a preempt call; preemption was a common trigger for the truncated
25 overlap-clearance problem.

Ovlp 10 Yellow shown below is 1.00s below the median in 2.18.1 (2.15.1: 3.00s vs median 4.00s; 2.18.1: 3.00s vs median 4.00s)

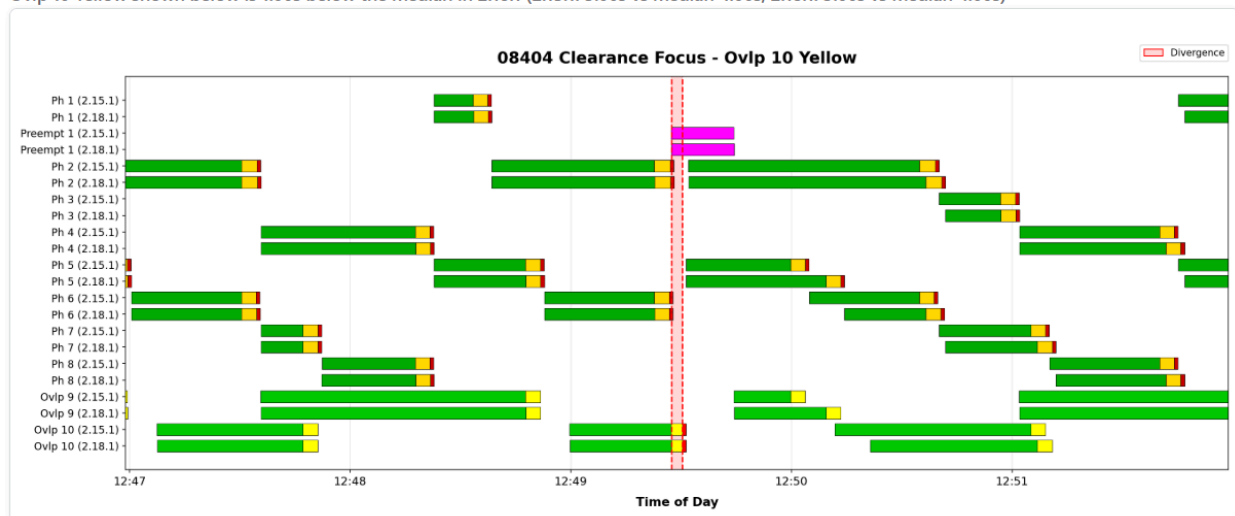


Figure 4: Truncated overlap yellow detected during preemption. The 3.0-second interval was below ODOT's 3.5-second minimum review threshold and the 4.0-second median yellow interval for this signal phase. This type of finding motivated the overlap trail-yellow and related clearance-setting changes.

1 Application to software-version acceptance

2 After repeatability was established, the method was applied to acceptance testing of a new software version. Config-
3 uration labels such as 12036 and 2B045 are ODOT internal tracking identifiers for individual signalized intersections.
4 Nineteen of 25 configurations passed the aggregate thresholds and six were directed to staff review. Mean sequence
5 match was 96.2 percent and mean timing match was 93.8 percent. Four configurations fell below the sequence
6 threshold; two exceeded the sequence threshold but fell below the timing threshold. These counts describe how the
7 review was organized; the important result is that the aligned traces exposed specific changes in signal operation.

Outcome	Count	Configurations / interpretation
Passed both thresholds	19	Operationally similar overall; detailed diagnostics still identified localized changes.
Sequence review	4	12036, 2B045, 2B094, 2B339. The 12036 result likely reflects peer-to-peer operation not being configured correctly in the test environment.
Timing review	2	2B054 and 2B085.

8 Table 4. MAXTIME version 2.15.1 versus version 2.18.1 release-comparison outcomes.

9 The detailed findings were more informative than the aggregate status:

- 10 • **Rail-preemption fix (13008).** Preempt 6 is a custom ODOT rail-preemption function that inhibits pedestrian service
11 while preparing for the main rail preempt. In MAXTIME version 2.15.1, Preempt 6 sometimes remained active after
12 its input ended. Version 2.18.1 exited the preempt correctly (Figure 5), confirming the known fix.
- 13 • **Transition-algorithm difference (2B049).** The aligned traces showed that the two versions used different short-
14 way coordination-transition logic. The difference did not appear to change signal operation. The trace also raised a
15 separate question: the controller appeared to be in step while still reporting transition (Figure 6).
- 16 • **Expected clearance behavior (13008).** The report flagged an irregular phase 4 red-clearance interval. Review
17 showed that dynamic red-clear extension was configured, and both versions handled it consistently. The flag
18 demonstrated that an unexpected clearance change would have been visible.

19 The first two configurations passed the aggregate thresholds. This is important: the tool did not merely label controllers
20 pass or fail. It retained localized differences that confirmed an operationally important bug fix and exposed an
21 apparently non-operational algorithm change within otherwise similar full-day operation.

Preempt 6 Active: largest local mismatch window; mismatch 790.40s; 2.15.1 events 1, active 790.40s; 2.18.1 events 0, active 0.00s

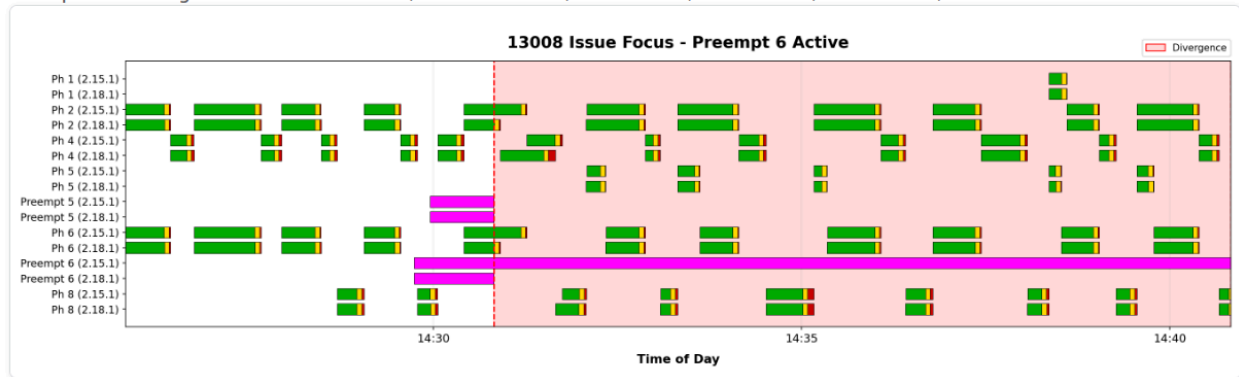


Figure 5: Configuration 13008, Preempt 6. The custom ODOT function inhibits pedestrian service in preparation for the main rail preempt. MAXTIME version 2.15.1 failed to exit after the input ended; version 2.18.1 exited correctly.

Transition Longway: largest local mismatch window; mismatch 917.50s; 2.15.1 events 1, active 686.40s; 2.18.1 events 1, active 231.10s; window avg 2.15.1 686.40s vs 2.18.1 231.10s

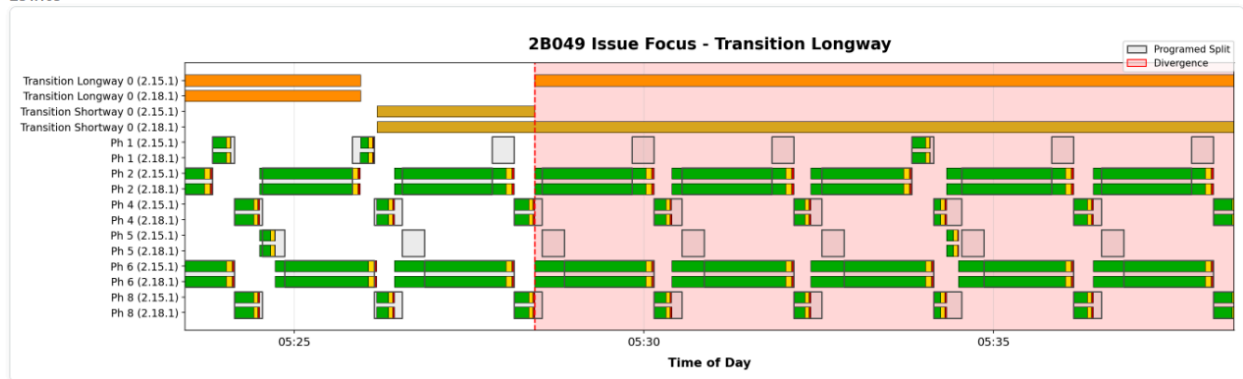


Figure 6: Configuration 2B049, short-way transition. The aligned traces show a change in transition logic between MAXTIME versions. No operational difference was apparent, although the controller appeared to be in step while still reporting transition.

- 1 No new operation-impacting software defect was confirmed in MAXTIME version 2.18.1 from the reviewed results.
- 2 The new version was otherwise operationally similar across the tested configurations, and the virtual conflict monitor
- 3 recorded no incompatible simultaneous outputs.

4 DISCUSSION

5 ODOT now uses the method for controller-software acceptance. It enhances controller testing; it does not simply
6 automate the same manual test. Staff can still check individual functions deliberately, while field replay adds approxi-
7 mately 23 hours of interacting inputs across 25 actual timing configurations. Available emulators run concurrently, and
8 the software checks millions of output events for sequence and timing changes, clearance irregularities, preemption and
9 pedestrian differences, and incompatible signal phases and overlaps. Staff can concentrate on the exceptions instead
10 of choosing and operating every input combination.

11 The high repeatability of the unchanged runs is what makes this practical. If repeated controller outputs could not
12 be aligned, an unexpected software effect would be indistinguishable from ordinary run-to-run variation. Instead, the
13 same inputs produced closely matching sequences, while deliberate timing changes and a software bug fix remained
14 visible. The comparison therefore provides a broad safety net for unforeseen changes in every behavior exercised by
15 the replay.

16 Field logs also capture combinations that are difficult to anticipate: simultaneous vehicle and pedestrian calls, time-of-
17 day plan changes, coordination transitions, and preemption during actual traffic operation. A rare failure or flash event
18 can be saved as a targeted replay, used to reproduce a reported problem, and then run again to test a vendor correction
19 or timing workaround.

1 The accepted baseline is a reference trace, not an assumption that the old operation is always correct. The 13008
2 preemption difference was desirable because the candidate software fixed a baseline defect. Staff must classify each
3 reported difference as expected, beneficial, harmful, caused by the test environment, or unexplained before accepting
4 a new baseline.

5 The open-source package separates NTCIP inputs from product-specific outputs. The replay-input method is not tied
6 to a controller brand and can be applied to controllers that implement the required NTCIP objects. MAXTIME is the
7 implementation evaluated here, not a limitation of the underlying replay method.

8 **LIMITATIONS**

9 The current event-log collector supports MAXTIME only. Supporting another controller family requires a new event-
10 log interface and event-code mapping. Replay capacity is limited by available computing and network resources,
11 but this evaluation ran as many as 15 emulators simultaneously on a standard Windows desktop. The 23-hour traces
12 cover only the conditions experienced by the 25 field controllers during the recorded day. An edge case occurring on
13 another day or at another configuration will be missed unless that trace and configuration are added. Current time
14 and computing constraints make testing every agency controller impractical, although additional complex or critical
15 intersections can be included as needed.

16 **CONCLUSIONS**

17 This study found that field-derived high-resolution event replay is feasible for comprehensive controller testing.
18 Independent unchanged runs averaged 99.6 percent sequence match and 97.7 percent timing match, demonstrating that
19 DTW could reliably align complete controller-output sequences. The comparison then flagged 11 of 14 configurations
20 with deliberate overlap clearance-setting changes and none of the unchanged configurations. Applied to a software
21 update, it surfaced real operational differences within otherwise similar full-day operation, including a confirmed rail-
22 preemption exit bug fix. The virtual conflict monitor found no incompatible signal phases or overlaps in the three
23 campaigns.

24 The significance is the increase in test coverage. Instead of relying only on staff to select and toggle individual
25 inputs, agencies can replay actual field activity across many production configurations and automatically compare the
26 resulting signal phases, clearances, pedestrian service, preemption, coordination, and conflicts. This creates a practical
27 opportunity to discover unforeseen software or timing effects in the test environment rather than after field deployment.
28 The same process can reproduce field failures and verify proposed fixes against identical inputs.

29 The open-source package provides a reusable NTCIP-based input interface and a comparison method that other
30 agencies can adapt. Controller-specific output collection and event mapping are still required, but the successful
31 alignment and detection results establish that the underlying approach works.

32 **ACKNOWLEDGMENTS**

33 The author thanks Chris Primm, State Traffic Operations Engineer, Oregon Department of Transportation, for
34 reviewing the results and contributing ideas for testing.

35 **CODE AND DATA AVAILABILITY**

36 Signal-Replay is an open-source Python package available from GitHub and PyPI. The study used repository version
37 0.2.0, which includes replay and comparison code, tests, documentation, and offline examples.

38 The repository includes a self-contained comparison-stage example for configuration 13008: saved MAXTIME 2.15.1
39 and 2.18.1 output events, comparison settings, expected results, and a reproduction script. Running the example regen-
40 erates the HTML report and verifies 98.6 percent sequence match and 96.6 percent timing match without controller
41 hardware, firmware, or configuration databases. It reproduces comparison and report generation only; the original
42 controller replay still requires controller/emulator software, configuration databases, and agency test infrastructure.

43 **GENERATIVE AI DISCLOSURE**

44 Generative artificial intelligence tools, including Claude Code and OpenAI Codex, were used to assist with software
45 development and review, data-analysis workflows, figure preparation, manuscript review, and document formatting.
46 The author reviewed and tested the software, verified the cited sources and reported results, and accepts responsibility
47 for the research methods, findings, interpretations, and final manuscript.

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